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Agents, Alignment, and the Many Faces of Autonomy

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Abstract
This article investigates the prospect of aligning AI agents with human autonomy. We show that doing so is a challenge: while there is general agreement on the concept's broad contours, its components can be specified in a number of conflicting ways, making it difficult to decide how an autonomy-enhancing agent should act. In response, we introduce three strategies for dealing with disagreements about interpretations of autonomy. Each offers a coherent approach that captures a different set of considerations, with its own strength and trade-offs. The liberal approach is committed to user integrity and promotes human autonomy by acting strictly on what a person says they want, without trying to change their mind. The capability-boosting approach is built around the idea of flourishing and promotes human autonomy by empowering the user to act on their goals. The meta-autonomy approach prioritizes choice by giving the user the kind of autonomy they want in relation to their interactions with an AI agent. By highlighting the multifaceted nature of human autonomy, this article contributes to recent efforts seeking to identify and operationalize targets of AI value alignment. It also illustrates the challenges and trade-offs that can arise not just between—but even within—the values selected as targets for alignment and offers strategies to navigate them.

Keywords AI agents · AI alignment · Autonomy · Boosting · Meta-autonomy · Meta preferences

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1 Introduction

Artificial intelligence (AI) systems are becoming more agentic. Capable of predicting, planning and executing actions, AI agents can increasingly perform tasks without human oversight (Knight, 2024; Russell & Norvig, 2016).[1] Their rapidly evolving capabilities are likely to be accompanied by ever-deepening integration into social contexts, with potentially far-ranging implications for individuals, groups, and communities.[2] This puts AI agents at the frontier of AI ethics (Gabriel et al., 2024; Lazar, 2024). Indeed, a large number of technology companies currently aim to develop personal AI agents—a type of general agent that fuses advanced technological capabilities with social and relational elements (Google DeepMind, 2024). Personal AI agents are explicitly designed to serve human needs, for example through augmenting or delegating cognitive tasks.[3]

As people enter increasingly interdependent relationships with AI agents, concerns about human disempowerment and loss of agency loom large (Kirk et al., 2025; Kosmyna et al., 2025; Kulveit et al., 2025). In particular, there is a worry that AI systems with enhanced capabilities will replace human participation across different domains and gradually erode people's ability to take charge of their own life. A precise articulation of what increased AI capability means for human autonomy, and how these worries might be addressed, is hard to find. But efforts to reverse these trends rapidly encounter a challenge that has long been noted in political and moral philosophy (Berlin, 1969; Carter, 2004; Christman, 2021): The idea of autonomy is itself multi-faceted, generating a range of tensions and potential contradictions. For example, does autonomy involve doing what one wants, even when those actions undermine one's long-term independence? Is a person suffering from addiction more autonomous when they are able to get what they want or when they are impeded from acting on their desire? These conundrums carry over into AI alignment: What should an aligned agent aim to achieve, and how do we deal with trade-offs between meeting a user's immediate desires and safeguarding their agency over time?

In this paper, we deal with these questions by examining how interactions between humans and AI agents affect autonomy. While there isn't a unified account of the concept, human autonomy can be broadly understood as a person's ability to act according to their own preferences or desires, and to do so in a way that is free from others' interference or undue influence. Previous research has discussed human autonomy in the context of users' willingness to accept algorithmic recommendations (Fink et al., 2024), the moral appropriateness of human-AI relationships (Kirk et al., 2025; Manzini et al., 2024), the legitimacy of automated forms of influence (Benn

[1] Russell Norvig (2016) characterize agentic AI by its ability to perceive its environment and act on it. For the purpose of this paper, we use the terms AI agents, agentic AI, and agentic algorithmic systems interchangeably (see also Chan et al., 2023).

[2] To date, agents are already used for writing code (Yang et al., 2024; Jimenez et al., 2024), economic transactions (Maslej et al., 2025), research (Google, 2025; Ramos et al., 2025), education and learning (Baha et al., 2023; Chen et al., 2022), creativity (Zhang et al., 2023), building relationships (Manzini et al., 2024), or providing therapeutic advice (Nicioli, 2024).

[3] This differentiates them from other types of agents, such as those used in logistics, trading, supply chain management or traffic control.

& Lazar, 2022), epistemic independence (Koralus, 2025) and socio-technical safety (Weidinger et al., 2023). Yet, research has not addressed the relationship between personal AI agents and human autonomy, and the methods to align them. In particular, we need to ask whether it is possible for a person to undergo deep and persistent interaction with a personal AI agent and remain as autonomous as—if not more autonomous than—they were before.

Against this background, this paper explores how AI agents can be designed to protect and promote human autonomy. We introduce the concept as one whose precise interpretation is contested. Instead of offering a single unified account, we formulate three coherent strategies that target different dimensions of autonomy, emphasizing user integrity (the liberal approach), flourishing (the capability boosting approach), and choice (the meta-autonomy approach), respectively. In doing so, we contribute to a growing field of research that investigates targets of value alignment (Gabriel & Keeling, 2025; Zhi-Xuan et al., 2024). By highlighting the many faces of autonomy, we also show that value alignment needs to navigate trade-offs not just across, but also within, individual values for alignment.

The article proceeds as follows. The first section introduces human autonomy as a contested concept. This fragmentation raises the challenge of how to align personal AI agents to human autonomy, and what interpretation an effective alignment strategy ought to prioritize. The second section illustrates this dilemma by exploring the multiplicity of human preferences. We show that conflicting and competing preferences create difficult trade-offs, forcing users and developers to make hard choices about what kind of autonomy AI systems should promote. The third section addresses this challenge by formulating three alignment approaches that advance human autonomy through non-interference, capability, and meta-autonomy, respectively. The fourth section explores broader implications for AI value alignment. We conclude by encouraging users, developers, and society to think concretely about the kind of autonomy they want to see realized in human-AI relations.

2 Personal Agents and the Anatomy of Autonomy

While there is scholarly disagreement about the specifics of human autonomy, it is often conceptualized in terms of a person's ability to act according to their own preferences or desires, and to do so in a way that is free from others' interference (Christman, 2009; Dworkin, 2012; Prunkl, 2024; Raz, 1988). Scholars have also highlighted that the quality of a person's thought and action is central to their status as an autonomous person (Calvo et al., 2020; Frankfurt, 1971). Indeed, a person can generally be said to be autonomous to the extent that they formed their preferences independently—nobody forced or manipulated them to want what they want—and to the extent they can do what they want without being impeded (Christman, 2009; Raz, 1988). In other words, autonomy is a form of self-governance (Christman, 2009; Dagger, 2005), built around three conceptual components: non-interference, prefer-

ences, and abilities.[4] We will now discuss these components in the context of personal AI agents.

Autonomy requires non-interference. Most scholars agree that an autonomous person needs to be free from others' interference with their agency. For instance, if Anna wants to drive to Los Angeles via the scenic Pacific Coast Highway, but her AI agent keeps interjecting her driving to reroute her toward the faster Interstate, Anna's autonomy may be reduced. Yet, interference can at times also be autonomy-enhancing: If the agent blocks Anna from typing text messages while she is driving, the agent might be thought to promote her autonomy. One way of understanding this distinction centers upon whether the interference is conducive or opposed to the person's long-term preferences, as well as the context in which the human-agent interaction occurs. Another, inspired by the intellectual tradition of republicanism, is to focus on the source of control (e.g., Pettit, 2012): Has Anna willingly yielded power to the AI agent, and does she have the means to regain control if she wants to?

Autonomy requires authentic preferences. Several accounts posit that being autonomous requires that a person has some kind of desires that can be assumed to be their own (Christman, 2009; Prunkl, 2024). On some interpretations, this is merely a procedural requirement, meaning that a person cannot act on wants that are the result of undue influence or manipulation. If Anna diligently acts on her preference to learn Spanish each day, but this preference was secretly instilled in her by her AI agent, Anna is less autonomous than if she acted on motives that were truly her own (Dworkin, 2012; Freyenhagen, 2015).[5] This differs from substantive views of autonomy, according to which a desire must be congruent with the person's standing to be authentic (Oshana, 2006; Benson, 2009; Stoljar, 2000).[6] Others, particularly feminist scholars, have pressed for a more relational view on autonomy that foregrounds people's social embeddedness and (power) relations that influence not just their view of themselves, but also processes of deliberation and reasoning (e.g., Hirschmann, 2003; Mackenzie & Stoljar, 2000; Meyers, 1987).

Autonomy requires the ability to act. To be autonomous, a number of accounts stress that a person must have the ability and opportunity to act on what they want. This can include having a range of meaningful choices available to them (Raz, 1988), or having access to capabilities to make progress toward their goals, even if achieving a specific outcome is not guaranteed (Nussbaum, 2011; Sen, 1979). In the first instance, the agent increases the user's autonomy by offering them a choice between valuable alternatives. An example of this is when Anna struggles with her statistics

[4] People can enjoy different degrees of autonomy both compared to others and within themselves (Dagger, 2005, p. 198; Christman, 1989, p. 13–14). Most people enjoy different degrees of autonomy in different contexts. While it is unrealistic for most people to ever be maximally autonomous, one can still act autonomously in specific contexts (Christman, 2013). It is this latter sense of autonomy that we are concerned with here.

[5] Yet, there is the question of how procedural neutrality is achieved. In addition to the absence of manipulation, some focus more on some sort of congruence between who a person is and what they desire (Christman, 2009; Dworkin, 2012).

[6] While intuitively plausible, such interpretations fail to incorporate common intuitions about autonomy: Anna can relocate because of her partner, voluntarily giving up some independence while still being autonomous (Dworkin, 2012).

assignment and the AI agent offers her a choice between summarizing existing course material, explaining the relevant concepts, or completing the assignment for her. In the second instance, the AI agent enhances Anna's autonomy by helping her develop the necessary capabilities, enabling her to do the assignment herself, for example through increasing her declarative or procedural knowledge: knowledge of facts, or knowledge about how to perform actions, respectively (Ten Berge & van Hezewijk, 1999).

What exactly these components denote, and how they fare against each other, is subject to further debate. For example, while some accounts emphasize the person's ability to choose among a range of meaningful options (Raz, 1988), others require the person to be able to effectively act on their goals (e.g., Christman, 2009; Prunkl, 2024). Similarly, there is disagreement as to what exactly makes a person's desire their own: does it merely require the absence of manipulation, or does the substance of the preference need to be congruent with the person's identity overall (Oshana, 2006, Stoljahr, 2000)?

This brief discussion has shown that while human autonomy possesses a conceptual core organized around non-interference, preferences, and abilities, it is also subject to different interpretations as to what these elements mean and how they should be specified. Addressing this multi-dimensionality in the context of AI agents requires making hard choices (Dobbe et al., 2021; Goodman, 2021). The next section deals with the multitude of preferences as an illustration of this complexity. We focus on a particular part of this challenge: understanding the types of human preference that could anchor autonomy, as well as the trade-offs that arise when we try to operationalize them for AI alignment.

3 Human Preferences: Stated, Revealed, Informed, and Ideal

Type of preference	General definition of autonomy	Definition of autonomy in an AI agent context	As a sentence
Stated preference	A person's autonomy is enhanced (and not impeded) whenever they can act freely on their stated preferences	A person's autonomy is enhanced (and not impeded) whenever the AI agent acts to further their stated preferences	I want what I say I want
Revealed preference	A person's autonomy is enhanced (and not impeded) whenever they can act freely on their revealed preferences	A person's autonomy is enhanced (and not impeded) whenever the AI agent acts to further their revealed preferences	I want what my actions reveal I want
Informed preference	A person's autonomy is enhanced (and not impeded) whenever they can act freely on preferences they do or would have if they had access to all the relevant information	A person's autonomy is enhanced (and not impeded) whenever the AI agent acts to further preferences that they do/would have if they had access to all the relevant information	I want what I would want if I had access to all the relevant information

Type of preference	General definition of autonomy	Definition of autonomy in an AI agent context	As a sentence
Ideal Preference	A person's autonomy is enhanced (and not impeded) whenever they can act freely on preferences that promote their objective interest	A person's autonomy is enhanced (and not impeded) whenever the AI agent acts to further preferences that promote their objective interest	I want what best promotes my objective interests

Human preferences are not just integral to human autonomy; they also play an important role within contemporary AI alignment, as they are considered to be an effective way of representing human values and are easily operationalizable (Warren, McGraw, & Van Boven, 2010; Lichtenstein & Slovic, 2012). This approach has its roots in rational choice theory and builds upon a model of the individual as a rational and well-informed actor that knows what is good for them and seeks to maximize expected utility (Zhi-Xuan et al., 2024). Yet, preferences are also context-specific, adaptive, and dynamic (Caroll et al., 2024; Franklin et al., 2022; Pettigrew, 2019). And they can be misinformed, performative, or misdirected (Arneson, 1985; Gabriel, 2020; Sen, 1985). Moreover, there are different kinds of preferences that can be measured in a variety of ways, which may lead to conflicting accounts of what a person wants or desires. In other words, each type of preference specifies different aspects of a person's autonomy.

3.1 Stated Preferences

Stated preferences are what a person says they want. Anna telling her agent that she wants to learn about Ada Lovelace's impact on computing is an example of a stated preference. Focusing on stated preferences is both compelling and pragmatic because they offer an intuitive measure for human interests, and they are operationally easy to implement in large systems. Stated preferences also capture the intuition that a person might be the best advocate for what they want or need, and uniquely well-placed from an epistemic point of view to speak to this matter (Berlin, 1969; Ziegler et al., 2020; Leike et al., 2018). Their tractability makes them a popular strategy in AI alignment; notably in Reinforcement Learning from Human Feedback (RLHF), which aligns agents with a reward model constructed from stated preferences (Bai et al., 2022; Ouyang et al., 2022).[7] Yet, aligning an agent with what a person says they want also presents a challenge: Are a person's stated intentions really an accurate reflection of their authentic desires? After all, people often say they want something and then do something different. As subjective indicators, they can reflect social conventions, norms, or individual delusions (Dworkin, 1981).

3.2 Revealed Preferences

Revealed preferences are what a person's behavior indicates they want. If Anna is trying to solve a difficult calculation, and the agent knows that Anna has previously

[7] In RLHF, people state their preferences on different AI outputs for the same prompt, and the collected data is used to build a reward model that rewards an AI agent for behaving in ways that are likely to meet a person's preferences.

searched for tips on getting better at math, her search history reveals that she wants to enhance her own abilities. Revealed preferences are popular in AI alignment, in methods such as Inverse Reinforcement Learning (IRL), where the model infers the reward function from observed behavior (Ng & Russell, 2000; Zhu et al., 2024).[8] Yet, this approach is vulnerable to questions of validity. Recommender systems, for example, prioritize content based on observed user behavior, often influencing those preferences in a self-reinforcing process (Schuster & Lazar, 2024; Lazar et al., 2024; Jargon, 2023; Milano et al., 2020).[9] Personal AI agents that promote autonomy via revealed preferences therefore face an epistemic challenge: They prioritize action over self-representation—engaging in a form of behaviorism (De Yong & Prey, 2022). The link between user behavior and authenticity is further complicated by the fact that preferences revealed in behavior could occur due to heuristics or habitual thinking (Tversky & Kahneman, 1974) or artificially constrained or manipulated digital environments (Zhou et al., 2019).

3.3 Informed Preferences

Informed preferences are what a person would endorse if they were rational and well-informed (Gabriel, 2020). They deal with the procedural conditions for autonomy: what a person would want after careful consideration and with access to good quality information. For this reason, developing and acting on informed preferences is often a hypothetical ideal—in reality, people often don't behave rationally, and lack both time and access to the information necessary to develop informed preferences (Banerjee & Duflo, 2011). They are also affected by a range of decision-making biases that impede rational deliberation. Personal AI agents could change this, for example by increasing a user's declarative or practical knowledge. Still, an AI agent that promotes informed preferences could yield controversial results. In theory, a person could rationally endorse something that is objectively bad for them, as long as this corresponds to a coherent set of desires (Quinn, 2012; Sparks & Wright, 2025). For example, Anna might want to learn how to function with less sleep, a preference she rationally endorses but which does not align with her well-being, objectively speaking. Developing a rational and informed set of preferences may also necessitate efforts to integrate and prioritize among different levels of preferences, including immediate preferences and higher order (or meta-) preferences—preferences about preferences (Ashton & Franklin, 2022).[10]

[8] This can include user engagement, such as views and clicks, but also conversational acts, such as a user articulating opinions, statements or questions.

[9] A documented instance of such a runaway feedback loop were teenage girls being exposed to, and subsequently internalizing, unrealistic beauty standards and body images on social media platforms (Pedalino & Camerini, 2022; Wells, Horwitz, & Seetharaman, 2021).

[10] In philosophy, meta-preferences are commonly referred to as second, or higher order desires, while immediate needs and wants are termed first-order interests (e.g., Taylor, 1985).

3.4 Ideal Preferences

Ideal preferences are what a person would prefer if they prioritized what is in their overall best interest. They correspond with a substantive perspective on autonomy: what a person would want if they were making choices that promote their well-being. To define the latter in the context of ideal preferences, we follow the gist of objective list theories, which posit that people's well-being is tied to a set of objectively desirable goods that include health, creativity, happiness, meaningful relationships, and long-term autonomy (Parfit, 1984; Ryan et al., 2013; Ryff, 1989).[11] This perspective is also reflected in capability theories, a body of work concerned with the conditions that allow individuals to achieve desired beings and doings, or "functionings" (Nussbaum, 2011; Sen, 1979). One upside to an agent promoting a user's ideal preference is that this would likely coincide with promoting their long-term ability to make autonomous decisions, such as when Anna's AI agent acts on an ideal preference for continuous learning (Hershfield, 2011). Yet, prioritizing ideal preferences assigned by an AI system also carries substantial risk, since it assumes that there is something that a person ought to want irrespective of their own position in the matter (Hayek, 1960/2011). Taken to the extreme, the user ceases to be the authority on what they truly prefer.[12]

4 Aligning Personal Agents with Human Autonomy: Three Approaches

People have different types of preferences, all of which can occur and conflict with each other. For instance, Anna may tell her AI agent to order a fast food take-out for lunch, while acting in a way that suggests a wish to eat more vegetables and fruit. Or she can, after reading about the health benefits of fasting, arrive at the conclusion that skipping lunch will best serve her health, when what would actually improve her well-being were combining a balanced diet with regular exercise. These examples illustrate that there isn't a single unified account of human autonomy that an AI agent should be aligned with, forcing us to contend with different challenges (Goodman, 2021; Dobbe, Gilbert, Mintz, 2019). They also raise the question of what (type of) preferences an autonomy-enhancing agent should promote.

To address this question, we present three strategies that target different dimensions of human autonomy. Each foregrounds a different aspect of the person—integrity, flourishing, and choice—and offers novel pathways to navigating the trade-offs between different interpretations of the concept. As is often the case with strategies

[11] With its focus on human flourishing and well-being, our account of objective interest differs from utilitarian perspectives that tend to emphasize utility maximization. As Gabriel (2020, p.11) points out, while there isn't a widely accepted universal definition of well-being, the extent of the disagreement is relatively small, and most people agree on core aspects of the concept, as indicated by objective list theories.

[12] It is along these lines that a person can be, as Jean-Jacques Rousseau famously claimed, "forced to be free" (Rousseau, 1762/2018). The expression refers to the paradoxical situation that occurs when the state promotes the "general will" of the people as a whole, possibly at the expense of the liberty of the individual.

emanating from a specific interpretation of a value, our approaches are not the only ones possible. Instead they represent distinct strategies for dealing with disagreements about the best interpretation of human autonomy, and enable comparative analysis of the unique advantages and trade-offs involved. Implicit to this discussion is a subtle but important insight: even a single value like autonomy is not a straightforward alignment target; it is a multifaceted concept we need to elaborate on before operationalizing it.

4.1 The Liberal Approach

The first alignment approach prioritizes user integrity and posits that the best way to align a personal AI agent to human autonomy is for it to act strictly in accordance with what a user says they want—their stated preferences—and to do so without attempting to change them, anticipate user needs, or "fill in the gaps". In other words, an aligned AI agent only acts on what a person says they want, refraining from unprompted personalization, persuasion, or inference from behavioral cues. By giving the user maximal control over the agent's actions and only having it act on explicitly articulated intentions, the AI agent serves as a transparent and predictable instrument of the user's will, prioritizing individual responsibility and non-interference over welfare. It also acts in line with the user's expectations about directionality and purpose, minimizing the risk of undue influence or manipulation on the part of the AI system.

The liberal approach fosters human autonomy in several ways. First, it respects and protects the user's epistemic sovereignty. It does so by directly following a user's articulated intentions instead of attempting to change them, refraining from manipulative practices and other types of undue influence (Mill, 1859/2011). In addition to adhering to the liberal maxim of non-interference, the AI agent can also help the user navigate challenging situations where their intentions may diverge from behavior. If Anna explicitly asks the AI agent to maintain her privacy online and prevent third parties from tracking her, it can help her navigate digital environments to enact these preferences. Second, the AI agent can also expand the user's range of feasible options by executing tasks on their behalf and according to their direct instructions. Third, the agent's unwavering adherence to stated preferences creates a predictable interaction framework for the user. This in turn cultivates a more robust sense of control and trust over the AI system's operations, allowing users to confidently integrate the AI agent into their personal and professional workflows.

Being strictly anti-paternalistic, the liberal approach posits that the AI agent should promote a user's stated preferences even when doing so reduces their well-being or undermines their autonomy over time. Moreover, this approach posits that an aligned AI agent should refrain from uninvited personalization, such as trying to foreshadow or infer what the user wants whenever direct instructions are absent. This requirement holds even if revealed preferences indicate an affinity the AI agent could easily incorporate, or if there are potentially more beneficial alternative courses of action unknown to the user. As a consequence, the agent neglects unexpressed needs and latent desires. The only reason an AI agent will refuse to act on a user's stated preferences is if acting on them would unequivocally inflict harm on others. This constraint

primarily serves to safeguard the well-being of others, ensuring that the exercise of one person's preferences does not negatively impinge upon individuals external to the user-AI interaction.[13] Other hard limits may include severe self-harm or illegal activities.[14]

A significant difficulty with this approach is that people often do not know what they want, or construct preferences spontaneously (Warren, McGraw, & Van Boven, 2010). This means that stated preferences can lack internal consistency over time. To resolve this ambiguity, the AI agent would have to explicitly request direction and clarification and directly engage the user, even at the risk of annoying or confusing them. Moreover, as the liberal approach refrains from personalization beyond stated preferences, its effectiveness hinges on the clarity and specificity of user instructions—the broader the statement, the higher the chances that the model will fail to accurately execute actions, or do so in a way that fails to meet user expectations.

Another challenge is that a user may give a lot of instructions to their agents, which could reduce their own skills and critical thinking faculties, gradually eroding their autonomy (Kosmyna et al., 2025; Kulveit et al., 2025; Marchal et al., 2026). Finally, people may use their personal agent to do things that undermine their own objective interests in the name of autonomy, such as reckless gambling. This is in line with the libertarian view that a free person must have the right to make mistakes (Hayek, 1960/2011). Thus, while an aligned agent might safeguard the user from harming others and engaging in illegal activity, there is no way to predict the full list of potential negative consequences under such an approach. This might pose a risk of reputational damage for companies relying on this approach.

4.2 The Capability-Boosting Approach

The second strategy prioritizes flourishing and posits that an AI agent best promotes human autonomy by enhancing the user's ability to act on their preferences over time. According to this view, an AI agent should promote a person's capability by increasing their declarative or procedural knowledge. Declarative knowledge refers to people's knowledge about facts, concepts, and principles that allow for understanding, reasoning, and informed decision-making. Procedural knowledge encompasses skills, techniques, and methodologies (Ten Berge & Van Hezewijk, 1999).[15] Enhancing people's capabilities also directly translates to expanded agency, providing them with the direct means to pursue their goals, as well as with enhanced

[13] From a classic liberal viewpoint, even this concession may be contested. While the user is under a duty not to harm others, it does not necessarily follow that the developer should take any option that may cause harm off the table. Philosopher John Stuart Mill, for instance, argued that it should be permissible to sell poison, because it can be used for good and evil. Thus, the potential for dangerous use is not sufficient for restricting an activity, only activities that only cause damage should be banned. (Mill 1859/2011).

[14] The implementation of such protections is in principle achievable in contemporary LLMs, notably through different techniques including red-teaming, system prompts, and application of content filters designed to identify and restrict harmful instructions or outputs.

[15] Our approach broadly fits into the capability framework advocated by philosophers Amartya Sen and Martha Nussbaum, particularly with their emphasis on offering individuals the ability to act according to their own version of the good life. That said, our understanding of capability as comprising procedural and declarative knowledge differs from their respective accounts.

pathways to acquire further resources, such as social capital or material assets. By offering users the opportunity to increase their skills and knowledge, this approach seeks to empower users to attain the intellectual and practical resources necessary to live a richer and more fulfilled life.

The capability-boosting approach promotes the user's autonomy by giving them the ability to enhance their effectiveness as an agent, and perhaps even develop new preferences through knowledge. This is analogous to an approach to behavioral change called "boosting" (Hertwig & Grüne-Yanoff, 2017). Boosting strategies focus on enhancing individuals' existing capabilities or developing new ones, thereby empowering them to make decisions that they themselves determine will lead to an increase in their own welfare (Hertwig & Grüne-Yanoff, 2017).[16] Boosts are transparent and explicit interventions that rely on voluntary user cooperation (Hertwig & Grüne-Yanoff, 2017, p. 982). This approach differs from nudging, where the AI agent seeks to subtly steer the user toward a predetermined behavioral outcome (Prabhu, 2018; Thaler & Sunstein, 2009).[17]

Under the capability-boosting approach, an aligned AI agent promotes human autonomy in a variety of ways. Perhaps the most important one is that boosting allows the user to determine what an increase in their own welfare looks like and equips them with the necessary skills to achieve this. In theory, the approach is agnostic toward the type of preference the user ends up pursuing, but in practice, more informed or ideal preferences comprise the most likely target. While a weaker version of this approach likely foresees solely boosting capabilities the user reflectively embraces, a stronger version may include an agent pro-actively suggesting strengthening capabilities linked to a person's objective interest, such as the ability to regulate one's emotions or financial literacy. This means that the capability-boosting approach offers considerable flexibility in its implementation, including in whether the agent is allowed to act in a paternalistic way or not (Floridi, 2016).

Under this approach, the aligned AI agent can also promote human autonomy through informing and explaining the tasks it performs on behalf of the user in a clear and accessible fashion, providing them with the necessary knowledge to acquire the skills to act on their intentions (Franklin, 2022). If Anna's behavior indicates interest in a healthy diet, her AI agent can educate her about what exactly makes a healthy lifestyle and how she can achieve it, boosting her declarative and procedural knowledge, instead of merely ordering balanced meals and scheduling regular exercise. These interactions can facilitate user learning, allowing individuals to understand the processes, reasoning, or steps involved in the agent's actions, thereby enhancing their own knowledge and ability to form preferences they reflectively endorse. Increased capability can also improve the user's ability to critically engage with the AI agent's decisions. To be effective, these explanations need to be thoughtfully designed and presented in a way that users find engaging, relevant, and useful. In addition, promoting autonomy through boosting may also counterbalance the risks inherent to the

[16] An example of a boost is an educational program designed to improve numeracy skills, which can subsequently enable individuals to make more informed personal financial decisions (Franklin et al., 2019).

[17] A prominent example of nudging is the use of default rules, such as automatically enrolling individuals into organ donation programs unless they opt out (Beraldo & Karpus, 2021).

short-term benefits a user would derive from delegating a task at the cost of diminishing personal capacities and the quality of their work—what critics have called "deskilling" (Bravermann, 1998; Sambasivan & Veeraraghavan, 2022).

The AI agent can also draw on the user's behavioral history, such as repeated requests for help with a certain task, to identify types of capabilities they value and want to develop. It can then utilize this information to proactively offer resources, learning pathways, and opportunities to boost these capacities, so that the user's development in these areas continues to progress. Alternatively, the AI agent identifies what capacities are required to act on a preference of the user, for example through conversations, and then empower them to develop these capacities themselves. For instance, if Anna has an informed preference to be a better communicator at work, the AI agent can help her identify skills required to act on this preference, such as learning about mediation and conflict management, and help her attain them. Just like boosting strategies, the capability-boosting approach seeks to promote autonomy by fostering the user's own ability to form and act on their preferences through transparent and voluntary cooperation.

There are challenges to this approach. To start with, empowering users to pursue their own versions of well-being does not necessarily mean promoting their ideal preferences. This tension between autonomy and well-being is shared with the liberal approach. But contrary to the latter, boosting comes with demands on a person's time, volition, energy, and cognitive resources—all of which might temper the effects of short-term whims. Thus, while this agency burden will often involve more friction than simple compliance with a request, it also increases the chances that a person will seek to attain capabilities they reflectively endorse.[18] A stronger version of the capability-boosting approach may also carry a risk of paternalism, particularly if the AI agent gets to decide what knowledge and skills may be most effective to enhance user flourishing. While this risk is not insignificant, it may be offset by its contribution to people's long-term autonomy and abilities. The key distinction is that capability-boosting requires voluntary user cooperation and active consent, whereas stronger forms of paternalism replace the user's independent judgment with what the user should want. Boosting tips into this stronger form of paternalism at the point where the agent begins overriding or bypassing explicit user preferences in favor of ideal ones, without seeking the user's endorsement or making its interventions transparent.

Additionally, enhancing user capability through explanations often involves a trade-off between accessibility and accuracy. Research on AI's conversational properties indicates that the act of explaining can lead users to trust the agent without adequate evaluation (Vorvoreanu et al., 2025). This could mean that users develop preferences based on wrong or partial information. Conversely, poorly designed

[18] That said, the extent of the agency burden may depend on the capability and context in question. For instance, an agent could also enhance a user's critical thinking faculties by providing accurate and trustworthy information in a high stakes, low resources environment.

explanations may result in information overload, impeding rather than facilitating user learning (Ferguson, Franklin & Lagnado 2022).[19]

4.3 The Meta-Autonomy Approach

Different people in different contexts may be drawn to different kinds of AI interactions: some might prefer one that is liberal and focuses on non-interference, while another boosts their capabilities or helps them act on their ideal preferences. Other interaction options also exist, for example, a system that aims to embed healthy defaults. How should we decide what option to pursue? One possibility is to promote human autonomy by giving each person the kind of AI agent experience they seek. This approach aims to give users meta-autonomy—the freedom to choose the kind of autonomy they want in a specific domain of the human-AI interaction, as expressed through different types of preferences. For example, Anna might want her AI agent to follow precise instructions when writing a birthday message. When it comes to her weekly work-out, by contrast, she may want the agent to suggest exercises that promote her well-being, irrespective of whether her behavior indicates a willingness to do them or not. An AI agent geared towards meta-autonomy would learn the difference between a person's preferences in different domains and tailor its response accordingly, acting on Anna's stated preferences in the former case, and her ideal preferences in the latter.

The meta-autonomy approach seeks to promote people's autonomy by allowing them to choose what kind of freedom they want from their agent across different domains of interaction. It does so by allowing users to specify different modes of agent behavior across domains, so that a person can exercise direct control over their AI agent in one area of life while granting them greater discretion in another.

The meta-autonomy approach enhances human autonomy in a way that is sensitive to both the user and the domain in question. This differentiates it from the liberal and the capability-boosting approach. For instance, while the liberal approach seeks to align the AI agent to a user's stated preferences, the meta-autonomy approach may include the AI agent promoting the user's stated, revealed, and ideal preferences in different contexts, so that they enjoy the kind of autonomy they desire in a given situation. This means that the meta-autonomy approach seeks to equip the AI agent with a more sophisticated understanding of user expectations about autonomy, moving beyond immediate reactions and towards discerning how the user wishes their diverse (and sometimes competing) preferences to be interpreted, balanced, and acted upon by the AI system.

How can an AI agent discern and prioritize among different, potentially conflicting, user preferences? Trying to learn a person's meta-preferences—preferences over their own preferences (Ashton & Franklin, 2022)—can be helpful for an AI agent that seeks to promote meta-autonomy. Meta-preferences are informative in this context

[19] The application of interpretability methods can be understood as interventions akin to boosts, when used to enhance an individual's competencies to interact with and assess artificial intelligence (Franklin 2022; Herzog & Franklin, 2024). By providing insights into the workings of AI systems these approaches seek to improve a user's understanding of how the system arrives at its outputs, allowing them to identify biases and errors.

because they can serve as a reliable indicator of an individual's more stable values and reflectively endorsed preferences, even when these are not directly reflected in their immediate actions and inclinations (George 2006; Deck & Schlesinger, 2014). Once the agent understands the user's meta-preferences—that is, what kind of interaction the user wants across different domains—it can prioritize among competing preferences accordingly. For example, if Anna indicates that she wants her AI agent to promote her ideal preferences when it comes to sleep—such as getting eight hours a night—the agent will reinforce actions consistent this objective, such as muting notifications after 10 pm or declining social events on weeknights.[20]

This approach can be operationalized through certain forms of dialogic interaction between the agent and user.[21] The agent can actively probe for the underlying reasons, values, or long-term goals motivating an instruction or repeated behavioral pattern, for instance by asking the user clarifying questions or inviting them to situate their requests in a broader context. Alternatively, the user may explicitly articulate their higher-order preferences for a specific AI agent interaction. If Anna plans to heavily rely on her agent to optimize her calendar and schedule meetings, she likely also has a sense of how she wants it to help: by inferring her mental load from behavioral cues, by overriding social expectations to protect a healthy work-life balance, or by following her direct orders. Crucially, the efficacy of such conversational tuning will hinge on, and reflect, the degree to which a user is aware of their meta-preferences in a given context.

What happens when a user has no strong preferences about what kind of autonomy their agent should promote? This concern reflects Sunstein's (2019) argument that people should be free to *not* choose, and should not be compelled to do so. In such cases, rather than insisting on guidance, an agent geared towards meta-autonomy can prompt the user to elicit a preference by offering different defaults. If the user still does not have a strong preference, the AI agent may fall back on a more general policy reflecting what users tend to prefer in that domain This would conserve the user's time and cognitive resources for domains in which the exercise of choice is more salient to them.

The meta-autonomy approach also faces certain challenges. First, a person may not initially know what kind of interaction they want from a relatively novel and unfamiliar technology, particularly in unpredictable contexts. Forming robust preferences about agent behaviour requires a sufficient understanding of the agent's capabilities and of how different modes of interaction are likely to affect one's goals and

[20] Promoting a person's meta-autonomy does not mean enacting meta-preferences across contexts. For example, the agent may follow a user's meta-preference for informed preferences when selecting a movie to watch. But this meta-preference only applies to user-agent interactions with respect to movie selections. When it comes to playing music, for example, the user may be less interested in consuming high-quality content and want the agent to act on their revealed preferences instead.

[21] Personal AI agents' multimodal properties invite this kind of conversational tuning. For example, the meta-autonomy approach might be initiated by an initial exchange between the user and agent, where the person gives instructions about how they want the agent to behave, similar to an intake interview at the start of a retreat tailored to individual needs and experiences. Over time, this dynamic interaction could mean that personal agents come to serve as "mindfulness machines"—enabling the user to identify meta-preferences they were not previously aware of, refine initially vague ones, or establish contextual conditions for their application.

well-being over time. Second, there is the question of when these preferences are elicited. Eliciting meta-preferences *ex post* has the advantage of grounding judgment in actual experience, but people may revise their preferences retroactively to rationalise past choices (Floridi, 2016). Eliciting them *ex ante*—before the agent has settled on promoting one set of preferences over another—captures a user's anticipated preferences without the distortion of hindsight, but remains vulnerable to people's tendency to mispredict their future reactions to different experiences (Loewenstein & Schkade, 1999).

Taken together, these considerations point toward the importance of good initial calibration, for example through early conversations about which domains the user considers most relevant to their autonomy, and about what kind of behavior they would like the AI agent to provide. Since people may not yet know what kind of autonomy they want from their agent, presenting them with a range of preset configurations can help scaffold these choices. The agent may additionally personalize its approach based on domains widely recognised as consequential for personal autonomy, such as decisions related to finances, career change, or personal relationships. That said, there is no straightforward solution to preference change, and some trade-offs will likely remain unavoidable (Caroll et al., 2024).[22]

5 Implications

Our analysis yields four implications: the multidimensionality of human autonomy as an alignment target, the promise of meta-autonomy, the design implications of the alignment approaches, and the importance of understanding the impact of personal AI agents on collective autonomy as a future research direction. We now address these points in turn.

5.1 The Many Faces of Autonomy

Since there are multiple conceptions of autonomy with no single correct interpretation, it is necessary to recognize and balance inherent trade-offs. Which approach works best hinges not only on how people see themselves, but also what kind of service they seek from their AI agent and what kind of risks to autonomy arise for a given domain. People who prefer the liberal approach might primarily be concerned about avoiding external control or manipulation, and ready to forgo situations where an agent could further support their well-being. Others might prefer an AI agent that functions more like a coach and are willing to sacrifice convenience for increased effectiveness and boosted skills. And then there will be those whose interest is best served by having their agent function more proactively still, anticipating and enact-

[22] Carroll et al. (2024) discuss a variety of approaches that deal with preference change, including constrained real-time reward (Caroll et al., 2024), initial reward (Everitt et al., 2021), natural shift reward (Carroll et al., 2021; Farquhar, Carey, & Everitt, 2022), or privileged reward functions (Evans et al., 2016; Yudkowsky, 2004), and show that each presents challenges, including being overly conservative, susceptible to unwanted AI influence, or requiring normative choices about what reward function is the 'right' target for alignment.

ing their preference across a range of domains, even though this involves less active control. This topography of preferences, profiles, and design choices means that we need to engage users and society in a conversation about what the technology looks like and how it works. Such efforts could branch out to participatory design (Delgado et al., 2023) and even democratic alignment which aims to resolve value trade-offs through public processes and discussion (OpenAI, 2024; The Collective Intelligence Project, 2023).

5.2 The Promise of Meta-Autonomy

Our analysis showed that approaches that focus on meta-autonomy have distinct advantages and are potentially significantly underexplored, as they give users control over the kind of action the AI agent exercises in different situations. However, people may not know what they want from AI agents, and we may need new processes to elicit these preferences in a reliable way. This can happen through conversational tuning, both initially and over the course of human-agent interactions. Initially, the agent may ask the user about their expectations for agent behavior, and what situations they consider particularly relevant to their autonomy. To prompt the user's reflection, the agent could present them with some scenarios commonly considered relevant for autonomy, such as changes in a person's career or giving advice for health, to see what relevance the user assigns to them, and how they expect the agent to behave. Over the course of multiple interactions, the agent might also occasionally present a user with new suggestions about their preferred mode of interaction based on past exchanges or requests and ask the user to verify. Over time, agents could raise users' (self-)awareness, getting to know themselves, their values and how these map onto certain kinds of agent interactions (Klingefjord, Lowe, & Edelman, 2024). Another possibility is presenting the user with a set of configurations from which they can choose based on their values and personal stance toward liberty versus self-realization.

Additionally, emerging technical work on value elicitation offers promising building blocks for operationalizing meta-autonomy. For instance, moral graph elicitation uses LLM-guided conversations to surface the values that inform people's choices and reconcile them into auditable alignment targets (Klingefjord et al., 2024). Complementary work on pluralistic alignment formalises frameworks for building AI systems that can reflect diverse user perspectives (Sorensen et al., 2024). And rapidly growing skill libraries specify instructions, resources, and scripts to create reliable workflows for AI agents (OpenAI Developers, 2026).

5.3 Plurality of Design Directions

The proposed alignment approaches point in different design directions for autonomy-promoting AI. To successfully operationalize the liberal approach, it is extremely important that we have agents that are maximally responsive to user intentions. Because the AI agent strictly adheres to a user's stated preferences, whatever is left unknown, ignored, or forgotten, remains disregarded. This requires that agents aligned this way need to understand implicature as they are not allowed to draw on other signals for alignment (Russell, 2019; Kasirzadeh, 2023). At the heart of this

is the capability to deal with ambiguity in the user's prompts and the context the agents are operating in. Second, to operationalize the capability-boosting approach, it is important to know what boosts human capabilities, and what atrophies them in human-agent contexts (Kimani et al., 2019) something that likely requires deeper investment in longitudinal studies of human-agent interaction (Fang et al., 2025). For the meta-autonomy approach to succeed, we need to develop robust forms of preference elicitation for agent behavior. Perhaps more importantly still is figuring out how an AI agent may calibrate its behavior across contexts. For example, a user might have a general preference for incorporating mindfulness exercises. But while this preference applies to many contexts—such as sleeping, eating, or driving—there might be exceptions, such as when the user is sitting in the car with someone else. To achieve a better calibration of agent behavior across contexts, agent personalization will likely be central.

5.4 Implications for Collective Autonomy

A number of concerns about the relationship between AI agents and human autonomy, particularly in the economic and political domain, are best understood not via the dyadic interaction of user and AI agent but rather at the collective level. This paper has focused primarily on one-to-one human-agent alignment, and its impact on autonomy. Yet, we also need to look at the multi-party question and explore what happens, from the standpoint of collective autonomy, when a large number of people use agents that are aligned to their individual preferences (Hammond et al., 2025). Is it possible that these individual autonomy enhancers will limit the agency of others, or affect the autonomy of a society as a whole in unexpected ways?

To answer these questions, future research should investigate AI agents' impact on collective autonomy. For instance, the impact of micro-level interactions on collective autonomy will likely be shaped by the type of individual level alignment that the majority of people or developers favor. If they choose an agent aligned with the liberal approach, we might see a lower risk of manipulation or externally induced preference change but greater risk that potential gains are left untapped when it comes to challenges such as education, health, or the climate. This differs quite substantially from a future where many sacrifice direct control or need satisfaction for long-term capability enhancement or the public interest.

Equally, as people's attitudes toward, and expectations for, their agents differ, we might observe a widening skills gap between different user groups if agent interactions are tailored to the meta-preferences of different users. That said, collective autonomy will likely also be shaped by the proliferation of these agents, their accessibility and affordability, in general, as well as the composition of the wider social, economic and political topography in which they are deployed (Fischli, 2024; Gabriel et al., 2024; Hammond et al., 2025). This will entail questions about social epistemology and trust in AI agents (Marchal et al., 2026). One criterion the current framework does not address is non-domination. Republican accounts of freedom focus on arbitrary interference as a distinct threat to freedom that cannot be reduced to preference satisfaction or capability (Dagger, 2005; Pettit, 2012). As AI agents become an essential part of the social and political fabric, they may also alter existing relations between

citizens and the state, and introduce novel or previously underappreciated risks to people's standing as free citizens. This paper focuses on users as private individuals, rather than citizens. Assessing our proposed alignment strategies against the standard of non-domination therefore falls outside the scope of this paper, but we flag it as an important direction for future research. Indeed, future research should dive deeper into how a "world of AI agents" (Gabriel et al., 2025) will impact society as a whole, as well as what effects human-agent interactions will produce at the collective level.

As AI agents become an essential part of the social and political fabric, they may also alter existing relations between citizens and the state—a concern this paper has not addressed. Here, the republican theory of freedom as non-domination can shed light on how AI agents may introduce novel or previously underappreciated risks to people's standing as free citizens, or how they might protect them against arbitrary interference from others (Dagger, 2005; Pettit, 2012).

6 Conclusion

As an alignment target, human autonomy is both desirable and challenging to operationalize. This has a lot to do with the fact that there is no unified interpretation of the concept, making it difficult to arrive at a single account that "fits" the full range of users and situations. Bearing this challenge in mind, we offer three strategies that highlight different ways of navigating autonomy-related trade-offs. Each offers its own perspective on how to harness personal agents' potential for human empowerment while minimizing their risks.

The liberal approach promotes human autonomy by prioritizing non-interference and foregrounding the user's integrity: an aligned AI agent enhances human autonomy by acting solely on a user's stated preferences. The capability boosting approach foregrounds human flourishing: it seeks to equip users with the knowledge and skills required to pursue their own version of well-being, often through the promotion of informed or ideal preferences. The meta-autonomy approach is perhaps our most novel contribution. It reframes the alignment challenge by emphasizing choice: users control what kind of autonomy is realized in their interactions with AI.

The importance of such efforts is underscored by recent developments. People are expected to form increasingly deep and dependent relationships with AI agents, as these are becoming both more capable and relationally embedded (Fang et al., 2025; Manzini et al., 2024). This raises novel opportunities and risks, especially in the context of situations where users are influenced by their agent in terms of how they understand their options or make decisions over time (Carroll et al., 2024; Kirk et al., 2025). The opportunities are clear: when aligned appropriately, personal agents can be autonomy enhancers, expanding people's agency and freedom in domains that matter. Yet, there are also risks inherent to increased AI agency, including concerns about gradual disempowerment (Kulveit et al., 2025) or a loss of human agency as a result of cognitive offloading, automation, and manipulation (Koralus, 2025; Mitelut, Smith, & Vamplew, 2023). In this paper we sought to make headway on these challenges, by providing a conceptual map and a research agenda for building agents that can genuinely enhance human autonomy.

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